

Adem Trabelsi

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A software engineering student with a strong interest in AI-powered applications and system architecture, Worked on projects integrating prediction and intelligent features, with a particular focus on transforming ideas into fully functional products, Driven by building software that is ready to ship, scale, and deliver value to real users.

SKILLS

Programming Languages

- **Frontend:** React, Angular, Flutter
- **Backend:** FastAPI, Node.js, NestJS,
- **Databases :** PostgresDB , MongoDB , Timescale DB SQL Server
- **Languages:** JavaScript, TypeScript, Python, C#
- **Tools:** Docker, Git

Data & Artificial Intelligence

- **Machine Learning:** Regression, Classification, Clustering , MLflow, Scikit-learn
- **Computer Vision:** Image processing, OCR
- **Deep Learning :** Basic deep learning models , Transformers , GANs
- **Data & AI tools:** Dataiku DSS , AI integration (chatbots, NLP)

PROFESSIONAL EXPERIENCE

KPMG - Tech Solutions - AI Intern

07/2024 – 07/2024 | Tunis

- Built a Smart Parking System with **computer vision-based vehicle detection**, **real-time database logging**, and a **live web dashboard for parking availability**.

Stack: Computer Vision · Image Processing · Real-time Database · Web Development

ULMUS Technologie - Mobile Application Developer

06/2025 – 02/2026 | Tunis

- Designed offline-first Flutter app for cash transport management; modeled SQLite schema for data consistency across rounds, passages, and collections.
- Implemented **secure bidirectional sync with error handling** and **automatic retry**; integrated **SUNMI V3 SDK for thermal receipt printing**.
- Conducted field testing with end users during cash collection rounds, **validating offline sync reliability** and refining UX based on real-world usage.

Stack: Flutter/Dart · SQLite · REST API · SUNMI V3 SDK · Offline-first Architecture

PROJECTS

Respiratory Sound Classification with Deep Learning (GAN Augmentation & AST-ASAM)

- Optimized an existing Contrastive Learning-based model for respiratory disease detection from audio signals, as part of a 3-person team.
- Implemented multiple **GANs** to address dataset imbalance.
- Designed data augmentation strategies to boost model performance using the **AST-ASAM framework**, achieving an overall score of **64%** (**61% Sensitivity, 67% Specificity**).

Real-Time Urban Carbon Emissions Monitoring & Prediction System

- Built a scalable Big Data pipeline using **Apache Spark, Kafka, and Hadoop** to process real-time air quality data from **853** stations across France. Integrated machine learning for **CO₂ forecasting and anomaly detection**, visualized through an interactive **Streamlit** dashboard with live charts and geospatial maps.

DeepGuard — Deepfake Detection System

- Built an end-to-end deepfake detection system with **Grad CAM explainability** using a **EfficientNetB0 + LSTM architecture** trained on **3,200 videos**, achieving **90% accuracy and AUC 0.956**.

FarmLink — AI-Powered Agricultural Platform

- FarmLink, an intelligent platform planned to empower farmers through AI and smart collaboration and help detect plant disease (**over 96% accuracy**)

EDUCATION

Software Engineering

2022 – Present | Tunis

National Institute of Technology and Science - INSAT

AWARDS AND CERTIFICATIONS

1st Place — National Re-Tech Fusion Hackathon (2026) - Industrial Energy Intelligence Platform

Built a multi-source energy data pipeline with CO₂ estimation and anomaly detection ; designed a 5-criteria waste heat recovery scoring framework ranking industrial heat sources across energy potential, CO₂ reduction, CAPEX, ROI, and integration complexity.

Dataiku Certified — Core Designer • Advanced Designer • ML Practitioner • Developer • MLOps Practitioner • Generative AI Practitioner

LANGUAGES

English — Fluent | French — Proficient | ARABIC — Native Language